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ELECTRICIAN JOB SHEET · PROJECT BASED

400 W automatic booster pump on battery backup: 70 m cable and Suntree ATS 2P 63 A at the pump

Harmony@Huailan Ecolodge, Chiang Mai · Job no. JOB-HHL-PUMP-01 · Issued by Solar ESS Working Group · 2 Nov 2026

How it is wired today

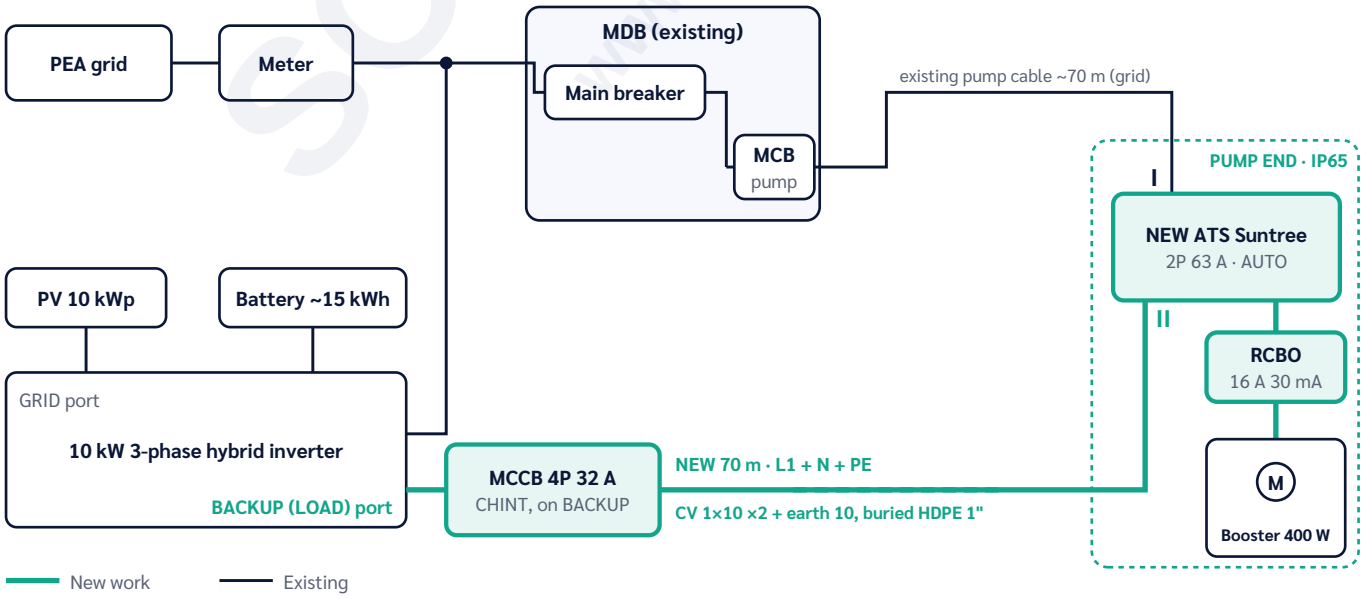
- PEA grid → meter → **inverter GRID port**, and from the same point to the **MDB → all loads**. The automatic booster pump (**400 W, single-phase**) is fed from the pump MCB in the MDB through an existing cable of about **70 m**. In a blackout the pump stops.

What we want

- Inverter BACKUP port → **CHINT MCCB 4P 32 A** (new) → take L1, N and PE → **new CV 1×10 mm² ×2 + 10 mm² earth, 70 m**, in buried HDPE pipe → **Suntree ATS 2P 63 A** input II at the pump.
- Existing pump cable (grid, from the pump MCB in the MDB) → ATS input I · ATS output → **RCBO 2P 16 A 30 mA** → pump (ATS and RCBO right next to the pump).
- ATS on **AUTO**: in a blackout it transfers to II (battery) by itself, and back to I when the grid returns.

Rules that must never be broken: never connect the BACKUP port in parallel with the grid. The ATS must switch both L and N (break-before-make). Never connect the inverter GRID port to the ATS output. Never join the BACKUP-port neutral to the MDB neutral bar.

Single-line diagram



The ATS and RCBO are right next to the pump. The ATS has two supplies: I = the existing pump cable from the MDB (grid), II = the new 70 m cable from the inverter (backup), protected only by the MCCB at its sending end.

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The new 70 m cable: size and type

Two checks decide the size. 1) Voltage drop for the 400 W pump (about 2.5 A running, 15 A starting): ≤ 3% running, ≤ about 10% starting. 2) Fault clearing: the only protection on this cable is the 32 A MCCB 70 m away, so a short circuit at the far end must draw more than its instant-trip level (about 10 × 32 A = 320 A). At 70 m, 10 mm² gives a good margin.

Copper size	Drop running / starting	Far-end fault current	Result
6 mm²	0.6% / 3.4%	≈ 395 A (> 320 A)	Passes, small margin
10 mm²	0.4% / 2.0%	≈ 570 A (> 320 A)	Passes – use this
16 mm²	0.2% / 1.3%	≈ 740 A	Passes, more than needed
10 mm² with 63 A MCCB	–	≈ 570 A (< 630 A)	Fails: that is why the breaker is 32 A

- **Cable type:** CV 0.6/1 kV (XLPE insulated, PVC sheathed), single core **1×10 mm²** × 2 (L, N), about 160 m in total, plus a **10 mm² green/yellow earth** (about 80 m) in the same pipe. The earth is the same size as the live cores, so an earth fault clears as fast as a short circuit. (NYY 3×10 mm² is an acceptable alternative.)
- **No THW in the ground**: use THW only for the short indoor runs.
- **Installation:** HDPE PE80 pipe 1 in (32 mm) buried ≥ 0.6 m (≥ 0.9 m under driveways), sand bedding, warning tape, pull boxes every ≤ 50 m and at bends, IMC risers. The new pipe may share the trench of the existing pump cable.
- **Existing pump cable (input I):** check it is rated for burial, at least 4 mm², insulation ≥ 1 MΩ, and protected by a 30 mA RCD in the MDB. If not, tell Solar ESS.

Protection

- **CHINT MCCB 4P 32 A** on the BACKUP port protects the whole 70 m cable. Not 63 A: the inverter maker specifies 32 A for this port, the port passes grid power through at up to 45 A only, and a 63 A MCCB (instant trip ≈ 630 A) would not trip fast for a fault at the pump end. Confirm the MCCB instant-trip setting is about 10 × In or lower. L2 and L3 stay free for other backup circuits.
- **RCBO 2P 16 A 30 mA Type A** next to the pump, after the ATS: protects the pump and gives shock protection there, on both supplies.
- **Suntree ATS 2P 63 A**: a switch may be rated above the breakers feeding it. Mount ATS and RCBO in one IP65 enclosure at the pump.
- Check that the inverter bonds N–PE in off-grid mode, so the RCBO also works on battery power.

Scope of work

1. **Survey:** pump nameplate (W, V, A), existing pump MCB/RCD and cable, the 70 m route, ATS position at the pump. Send to Solar ESS before buying materials.
2. **Isolate (LOTO):** grid main, inverter GRID/BACKUP AC breakers, inverter, battery and PV DC isolators. Prove dead.
3. **Supply end:** install the CHINT MCCB 4P 32 A on the BACKUP port in an IP54 enclosure next to the inverter; take L1 and N (plus PE) to the new cable.
4. **70 m cable:** dig the trench, lay HDPE ≥ 0.6 m deep, pull CV 1×10 mm² ×2 + 10 mm² earth, pull boxes every ≤ 50 m, IMC risers, backfill with warning tape.
5. **Pump end:** install the Suntree ATS 2P 63 A in an IP65 enclosure with DIN rail and terminals. Input I (N, L1) = existing pump cable (grid); input II (N, L1) = new cable (backup); output → RCBO 2P 16 A 30 mA → pump, all in one enclosure next to the pump. Set AUTO.
6. **Earthing:** join the PE of both cables and the pump in the ATS enclosure; an earth rod 5/8" × 2.4 m at the pump is recommended.
7. **Labels:** at the ATS enclosure and the MDB: "Pump fed from two supplies – live from the battery during a grid outage. Switch off both before work", and mark the buried route.

8 **Test and hand over** per the test table, with photos, an as-built diagram and the Solar ESS report form.

Danger: the ATS enclosure at the pump has two supplies, and the BACKUP port stays live from the battery even when the grid is off. Before work, switch off the pump MCB in the MDB and the MCCB at the inverter, lock out and tag out (LOTO), and prove both supplies dead. Locate existing water pipes and buried cables before digging.

Materials (minimum specification)

Premium materials only. Give the brand and size of every item in your quote.

#	Item	Minimum specification	Qty
1	MCCB on BACKUP port	CHINT MCCB 4P 32 A (63 A frame), 400 VAC	1
2	RCBO at the pump	RCBO 2P 16 A 30 mA Type A 6 kA	1
3	Supply-end enclosure	Steel IP54 with DIN rail and terminals	1
4	ATS	Suntree ATS 2P 63 A 230 VAC, DIN-rail, AUTO/MANU switch, I / II / Load ON indicators (or equal)	1
5	ATS enclosure at pump	Outdoor IP65 with DIN rail, terminals, earth bar, cable glands	1
6	70 m cable	CV 0.6/1 kV 1×10 mm² copper (L, N) + 10 mm² green/yellow earth	≈ 160 m + 80 m
7	Underground pipe	HDPE PE80 1 in (32 mm), sand bedding, warning tape; pull boxes every ≤ 50 m; IMC risers	≈ 70 m
8	Short indoor cables	THW 6 mm² in ¾" IMC/EMT	As measured
9	At the pump	Waterproof flexible conduit, earth rod 5/8" × 2.4 m with earth wire	1 set
10	Consumables	Lugs, ferrules, cable markers, warning labels, cable ties, anchors	1 set

Tests (record every reading)

#	Test	Pass when
1	Insulation of the new 70 m cable and the existing pump cable at 500 VDC, before and after backfill	≥ 1 MΩ
2	Resistance of each core of the new cable and PE continuity to the pump body	≈ 0.13 Ω per core
3	Earth rod resistance at the pump	≤ 5 Ω
4	Grid normal: ATS on I, pump cycles on its pressure switch; record voltage at the pump running and starting	≥ 207 V
5	Simulated blackout: open the grid main; ATS transfers to II; open a tap so the pump starts 5 times on battery; record voltage and starting current	No trip, ≥ 207 V
6	Grid back: ATS returns to I by itself	Automatic return
7	RCBO: test button and RCD tester (also on battery power)	Trips ≤ 300 ms

To confirm on site

- Pump nameplate: 400 W, single-phase 230 V, current (if above 1.1 kW, tell Solar ESS).
- Existing pump cable: type, size, condition, and whether it has a 30 mA RCD.
- The 70 m route: driveways, soil or rock, whether the existing trench can be reused.
- The inverter's off-grid N–PE setting.

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How to quote

- **Quote two prices:** 1) Labour cost only; 2) Labour & materials (list every material by brand and size).
- Price **trenching and backfill** separately per metre (normal soil / hard soil or rock).
- Pay guide for labour: **฿5,000** for 2–3 people, about 2 days including trench, cable pull, both enclosures and tests; materials and travel not included.
- Attach your electrician certificates and photos of similar work. Do not change any inverter settings; Solar ESS sets the inverter remotely.
- Send your quote with the form on www.solaress.org/job.html.

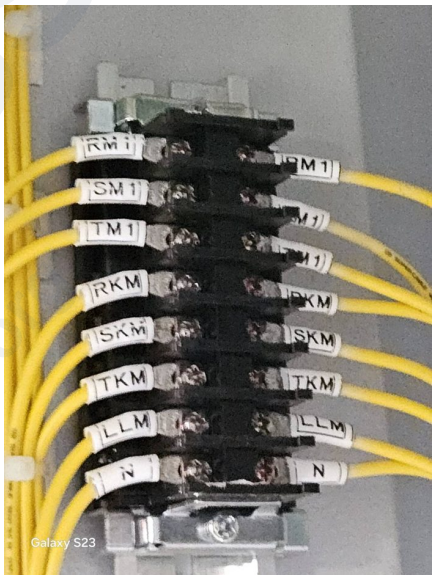
Site photos



Existing MDB (pump MCB)



Inverter room (MCCB 4P 32 A goes here)



Existing terminal block (identify before work)

Qualifications and certificates of the Solar ESS team are furnished upon request or upon completion of contract.

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